

Semantic prior

Method 3: Semantic Prior

1. Plan

Use semantic building information to condition DAv2 relative depth: $AGL = \beta_0 + \beta_1 D + \beta_2 P_b + \beta_3 (D \times P_b)$

where D = DAv2 depth and P_b = building probability/fraction.

Two separate roles:

- DFC2019 benchmark: derive P_b from RGB using an off-the-shelf building segmenter; never use DFC CLS as an input.
- Sentinel-2 India demo: use Microsoft GlobalMLBuildingFootprints, rasterized to the Sentinel grid with building fraction, 50 m density, edge and distance-to-building features. These are explicitly treated as an external semantic prior, not ground truth.

2. Data / models / paths

Sentinel RGB:

```
data/sentinel2/<city>/<City>_RGB.tif
```

GlobalMLBuildingFootprints:

```
data/sentinel2/semantic_sources/  
globalml_building_footprints/raw/
```

Input manifest:

```
data/sentinel2/semantic_sources/  
globalml_building_footprints/method3_input_manifest.json
```

The manifest records 5× supersampling, derived features, and the HOTOSM model path.

HOTOSM DINOv3-S building model:

```
models/semantic/hotosm_dinov3s_buildings/model.onnx
```

DFC RGB:

```
data/dfc2019/raw/RGB/Track1-RGB/
```

DFC AGL truth:

```
data/dfc2019/raw/Truth/Track1-Truth/*_AGL.tif
```

3. Sentinel semantic preprocessing + exclusion

Generated QC: [data/sentinel2/method3_semantic_QC.pdf](#) (contains mock numerical values, don't refer to values, refer to generated stats files in the mentioned folders instead)

We tested 10 AOIs. After QC, excluded: Kolkata, Delhi, Mumbai, Kochi because of large/inconsistent footprint coverage gaps.

Retained: Bardhaman, Bengaluru, Hyderabad, Jaipur, Darjeeling, Sundarbans (Sundarbans was retained as a useful near-zero-building case.)

The QC confirms the GlobalMLBuildingFootprints source and 2026-08-13 release; e.g. Jaipur has mean building fraction 0.2058 and p95 0.8674.

4. DFC semantic prior generation

Generated building-probability maps for the 50-tile DAv2 benchmark, saved as:

```
data/dfc2019/experiments/semantic/building/<tile>.npz
```

Example:

```
data/dfc2019/experiments/semantic/building/JAX_004_006.npz
```

5. Test 1 — same-tile regression

Result:

```
data/dfc2019/experiments/semantic/method3_results.json
```

50 tiles:

Metric	Baseline	Method 3
MAE	3.0349 m	2.8975 m
RMSE	4.2933 m	4.1231 m
Pearson	0.5457	0.5843

Spearman	0.4775	0.4901
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This showed improvement, but the regression was fit and evaluated on the same spatial tile regions.

6. Test 2 — spatial holdout

4-fold spatial CV: each tile was divided into four quadrants; three trained the model and the fourth was held out, repeated for all four folds. 50 tiles / 200 folds.

Result:

data/dfc2019/experiments/semantic/
method3_spatial_cv_results.json

Metric	Baseline	Method 3
MAE	3.3924 m	3.3948 m
RMSE	4.5787 m	4.9065 m
Pearson	0.5824	0.5542
Spearman	0.5093	0.4856

7. Conclusion

Method 3 is completed and rejected as the production correction method.

The semantic interaction model improves same-tile/in-sample metrics, but fails spatial generalization: MAE is essentially unchanged while RMSE, Pearson and Spearman worsen under spatial holdout.

Therefore:

Method 3 → retained as an ablation

Production path → move to Method 4